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Geometry-First Physics and the Protected Stratum

A constructive synthesis: how four-dimensional geometry fixes physics, and the stiffness that protects it

*Core document. Constructive treatment: the main text presents what the program **builds**. Every claim is tagged by status — **[proved]**, **[established (others’)]**, **[conjecture]**, **[interpretation]**. Routes considered and set aside are catalogued in the **Glossary** (§8). Provenance to the 25 working phases is in the Appendix.*

Abstract

We develop a **geometry-first** thesis: that the physical world is, at bottom, the geometry of a single four-dimensional manifold — with forces, quantum structure, matter, and even time *derived from* that geometry. The constructive core is that geometry fixes a great deal on a **protected stratum**: the gauge group is the Tannaka dual of the local-algebra net **[established]**; charge quantization is the integrality of a degree-2 cohomology class — Kaluza–Klein charge quantization *without the extra dimension* **[established]**; superselection sectors and the universal (topological) term of entanglement entropy are read off the *same* geometry-derived sector category **[established]**; and the entire low-energy effective dynamics is the special geometry of the Seiberg–Witten spectral curve, which is at once an integrable system and (by Taubes) the *J*-holomorphic-curve data of the symplectic 4-geometry **[established]**. Exotic smoothness is physical — it sources gravity

[proved]. Unifying these, we advance the program’s central conjecture, **Plank IV**: that the *protecting structure* — what lets geometry fix physics at all — is the **stiffness** (“folding rigidity”) **of spacetime**, its Godbillon–Vey invariant; spacetime can be folded (foliated) only so far before it becomes rigid, and **that rigidity is both the protection and, through its modular flow, the clock that generates time** — yielding the observed 3+1 split and the arrow of time from one structure [conjecture]. We give the constructive correspondence (monopole Floer homology = the protected superselection content, proved at the label level and on a controlled family) and the forward program. This is a **synthesis of established results** organized around new framings; the candidate-novel content (Plank IV foremost) awaits the scholarship check of §6. Alternatives reviewed and eliminated are listed in the Glossary (§8).

1. The thesis — four planks

The geometry-first bet is a set of **planks** (conjectures by nature):

- **Plank I — geometry is fundamental, and 4-dimensional.** The smooth 4-manifold — its metric and, crucially, its *diffeomorphism type* — is ontologically primitive; dimension 4 is *necessary*, because 4 is the unique dimension whose smooth structure is rich enough (exotic smoothness) to carry what others outsource to extra dimensions.
- **Plank II — quantum structure is geometry’s shadow.** Entanglement and the operator/Hilbert structure are the operator-algebraic shadow of the geometry (the AQFT net).
- **Plank III — forces are geometry.** The gauge structure is read from the geometric data, not added independently.
- **Plank IV — the protecting structure is spacetime’s *stiffness* (“folding rigidity”).** [central new conjecture] What lets geometry fix physics at all is a *protecting structure*; the proposal is that it is the **stiffness of spacetime** — its Godbillon–Vey folding rigidity. Folding (foliating) spacetime is possible only so far before it becomes rigid (like paper); any nonzero fold makes it rigid (a type III von Neumann factor), and **that rigidity is the protection** — and, via its modular flow, the **clock** that generates time.

The thesis, in one line: *a single four-dimensional geometry — whose stiffness both protects its physics and generates its time — is, with no extra dimensions and no fundamental supersymmetry, the substrate of physics.* What follows is what this picture **builds**.

2. What geometry fixes (the constructive results)

2.1 The gauge group, from the net [established]

The compact gauge group is the **Tannaka dual of the category of superselection sectors** of the local-algebra net (Doplicher–Roberts). On $M = \mathbb{R} \times \Sigma$ the abelian magnetic sector category is $\text{Rep}(\widehat{G})$ with $\widehat{G} =$ Pontryagin dual of $H^2(\Sigma; \mathbb{Z})$ [proved here as a corollary of **DR + Pontryagin duality**]. The gauge group is thus a functional of the geometry, exactly as Plank III asks.

2.2 Charge quantization, from cohomology [established]

Magnetic charge is the first Chern class, $g = c_1 \in H^2(\Sigma; \mathbb{Z})$ — Dirac quantization as the integrality of a characteristic class; via compactness this quantizes electric charge too. This is **Kaluza–Klein**

charge quantization without the extra dimension: the quantizing integer is a Chern number of the 4-geometry, not a momentum on a fifth dimension.

2.3 New charges from topology [established]

Nontrivial $\pi_1(\Sigma)$ produces Aharonov–Bohm superselection sectors (representations of π_1); non-abelian π_1 yields non-abelian sectors. Charge *content* is generated by spacetime topology.

2.4 Quantum structure as geometry’s shadow [established]

The net of local (type III₁) algebras fixes the factorization into subsystems and, by Reeh–Schlieder, the vacuum’s entanglement. The **universal (topological) term of entanglement entropy**, $\gamma = \log \mathcal{D}$, is fixed by the superselection-sector category — *the same category §2.1 derives from the geometry* (Casini–Huerta–Magán–Pontello + Kitaev–Preskill). Worked example: $S^2 \times S^1 \Rightarrow \widehat{G} = U(1)$, universal coefficient -1 . So the universal entanglement is geometric, and it is *the same object* as the gauge sector — Planks II and III meet here.

The bulk (area-law) term is also geometric — carried by stiffness, not by local fields [established mechanism + interpretation]. Entanglement entropy splits as $S = \alpha \text{Area} - \gamma + \dots$. One must separate the *correlations* from the *magnitude*: Bell/Tsirelson forbids local fields carrying the *correlations* (the carrier stays the type III₁ net, Reeh–Schlieder), but it does **not** forbid the area-law *coefficient* from being geometric — and it is. By **Susskind–Uglum**, the UV-divergent area term *renormalizes Newton’s constant*; the physical area-law that remains is **Bekenstein–Hawking**, $S = \text{Area}/4G$ — an area measured in units of the **gravitational stiffness** $1/G$. Since $1/G$ is the gravitational/Planck-scale reading of Plank IV’s stiffness (§4, §5.1), **both** terms of the entanglement entropy are carried by stiffness: the universal term by the *topological* (GV/sector) stiffness, the bulk term by the *gravitational* stiffness. The bulk entanglement is carried by **the stiffness of spacetime itself**, not by local geometric fields. (*That the two stiffnesses — GV and $1/G$ — are the same is conjecture; that each term is geometric via a stiffness is established.*)

This also supplies a candidate dynamical mechanism (cf. §6.1) — via an established program that must be credited. The variational principle Plank IV needs is an **entanglement-equilibrium / first-law** condition, and this is a decade-deep, much-cited body of work, *not* original here: - **Jacobson** (entanglement equilibrium): stationarity of vacuum entanglement entropy at fixed volume holds *iff* the (nonlinear) Einstein equation holds — intrinsic, semiclassical. - **Faulkner–Guica–Hartman–Myers–Van Raamsdonk** (gravitation from entanglement): the **entanglement first law** $\delta S = \delta \langle H_{\text{mod}} \rangle$ for all ball regions is equivalent to the **linearized Einstein equations** — the precise version (later extended to nonlinear). *Caveat: holographic — it derives the bulk equations from a boundary CFT, i.e. emergent geometry, which is in tension with Plank I (geometry fundamental). The Jacobson/Casini intrinsic versions are the Plank-I-compatible ones to lean on.* - **Casini** (the QFT-rigorous backbone): relative entropy, the modular Hamiltonian, and the Bekenstein bound from relative-entropy positivity make these statements precise in flat-space QFT.

The fit with the framework is exact: the **modular Hamiltonian** H_{mod} in the first law *is* the generator of the modular flow = thermal time = Plank IV’s engine. So the candidate derivation is **GV stiffness $\rightarrow$ modular flow (H_{mod}) $\rightarrow$ first law $\delta S = \delta \langle H_{\text{mod}} \rangle \rightarrow$ (linearized) Einstein equations**. What this program supplies — that equilibrium/first-law *derives* the dynamics — is *theirs* (established); what would be new here is only the identification of H_{mod} ’s modular flow with the **Godbillon–Vey stiffness** of the exotic foliation (see §6 novelty).

2.5 Low-energy dynamics, from special geometry [established]

The entire low-energy effective action is the holomorphic **prepotential** F , read off the **Seiberg–Witten spectral curve** ($\tau = \partial^2 F$, the moduli metric, BPS masses, couplings). That curve is simultaneously (i) the spectral curve of an **integrable system** (Donagi–Witten) and (ii), by **Taubes’** $SW = Gr$, the **J -holomorphic-curve data of the symplectic 4-geometry**. One geometric object — the curve — is the gauge group’s source, the integrable structure, the exotic-smoothness detector, and the law of the low-energy dynamics.

2.6 Exotic smoothness is physical [proved]

Exotic smoothness **sources gravity** (Brans conjecture; proved for compact manifolds by Asselmeyer-Maluga and for exotic $\mathbb{R}^4$ by Śladowski): it curves spacetime like matter, even in vacuum. The smooth structure — Plank I’s primitive datum — does physical work.

3. The unifying principle: the protected stratum

Across §2, geometry fixes physics on a **protected stratum**, and where it fixes *dynamics* it does so through a **protecting structure**. The master statement:

Geometry fixes physics to exactly the extent that a protecting structure is present — supersymmetry, holomorphy, integrability, or modular/thermal-time. The choice of protecting structure is the program’s only freedom; everything geometry achieves, it achieves on the protected stratum.

This is why the planks cohere: the sector category, the gauge group, the universal entanglement, the spectral curve, and (Plank IV) the stiffness are one structure seen from different sides.

4. Plank IV: stiffness as the protecting structure — and the origin of time

4.1 Stiffness = folding rigidity = the protection [conjecture; mapping established]

The protecting structure is identified physically as the **stiffness of spacetime**, by the paper-folding intuition: a sheet folds only so far before it is rigid. Precisely, in the live channel: - “*Folding into layers*” = a **codimension-1 foliation** (a stack of leaves) — the structure defining the exotic $\mathbb{R}^4$ (radial family $\leftrightarrow$ foliations of S^3). - “*Amount folded / stiffness*” = the **Godbillon–Vey invariant**, which Thurston interpreted as the “**helical wobble**” of a **pyramid of stacked discs** (Reinhart–Wood: via leaf curvatures); it varies continuously (uncountably many values). - “*Becomes rigid; the protection*” = $GV \neq 0 \Rightarrow$ no invariant transverse measure $\Rightarrow$ **type III von Neumann factor** $\Rightarrow$ a canonical **modular (thermal-time) flow** = the protecting structure of §3.

So **folding = foliation**, **stiffness = Godbillon–Vey**, **rigidity = type III**, **protection = thermal-time**. Connes’ theorem makes this canonical: the *flow of weights* — the state-independent invariant of the factor — *is determined by the GV class*, so the protection is geometrically pinned, not arbitrary.

4.2 Time is what the stiffness generates [interpretation]

In this picture the four dimensions are all geometric (the folded Euclidean 4-geometry); **Lorentzian time is emergent** — the modular flow driven by the GV stiffness (Connes’ flow of weights; Connes–Rovelli thermal time). The stiffness does not merely protect; it is the **clock**. (*This refines Plank I*

from “fundamental Lorentzian tetrad” to “fundamental Euclidean 4-geometry + GV stiffness, with time emergent.”)

4.3 The 3+1 split and the arrow of time, from one structure [conjecture]

- A codim-1 foliation slices the 4-geometry into **3-dimensional leaves** (the perceived spatial world) plus one **transverse direction**. We are embedded *in a leaf* $\Rightarrow$ we navigate its 3 directions as “space”; the transverse (folding) direction is where the modular flow runs $\Rightarrow$ we experience it *flowing* as “time.”
- **Why 3, not 2+2:** Godbillon–Vey is a **codimension-1** invariant; codim-1 in 4D forces **3 leaf + 1 transverse**. If the stiffness is GV, the world *must* present as 3+1.
- **Arrow of time, free:** the modular (KMS) flow is intrinsically **directional**, so emergent time is *one-way*. The 3+1 split *and* time’s one-wayness come from the *same* structure — where standard physics must add the arrow as a separate boundary condition.

4.4 Why 4D is the special dimension [established + interpretation]

The paper-folding *limit* is a real theorem — **Cheeger finiteness** (bounded curvature + diameter + volume $\Rightarrow$ finitely many diffeomorphism types). But **dimension 4 is the exception** (the clean finiteness holds for $n \neq 3, 4$; 4-manifolds admit *uncountably many* exotic structures). So 4D is the one dimension that **folds without a finite limit** — which is precisely why it is exotic, and why Plank I singles it out. The richness Plank I prizes and the stiffness/folding of Plank IV are the same fact about dimension four.

5. The constructive core: monopole Floer = the protected superselection content

The clearest single construction unifying the planks. On globally hyperbolic $\mathbb{R} \times \Sigma$ for the abelian Seiberg–Witten (monopole) theory: - **Label theorem [proved]:** magnetic superselection sectors $\cong \text{Spin}^c(\Sigma) \cong H^2(\Sigma; \mathbb{Z}) =$ the grading set of monopole Floer homology $HM(\Sigma)$. - **Internal dictionary [conjecture]:** $HM(\Sigma)$ is the protected content of each sector — the reducible/ U -tower is the global- $U(1)$ charge ladder, the Frøyshov invariant is the sector’s ground-state grading, irreducibles are charged-matter bound states. - **The bridge [proved on the protected sector, PSC $\cup$ all Brieskorn spheres]:** via Witten–Morse ($HM =$ SUSY ground states), the topological twist ($HM =$ the TQFT Hilbert space on Σ), and Osterwalder–Schrader positivity on the protected sector. Tested on S^3 (confirmed, and the dictionary sharpened) and all Brieskorn homology spheres (forced by isolated/nondegenerate critical sets). - **Stiffness reading [conjecture]:** the relevant Σ -data is exactly the Godbillon–Vey/foliation stiffness of Plank IV; $HM(\Sigma)$ is its protected, computable face.

This is the program’s most developed result: a proved label-level correspondence plus an internal correspondence confirmed clause-by-clause on a controlled family, all carried by the same stiffness/spectral-curve object.

6. Status and the forward program

What completes each result (constructive open problems), in order of leverage:

1. **Plank IV $\rightarrow$ mechanism (the decisive step).** Promote the stiffness identification from interpretation to mechanism. The variational principle is an **entanglement-equilibrium**

/ **first-law** condition — and that mechanism already exists (Jacobson; Faulkner–Guica–Hartman–Myers–Van Raamsdonk; Casini’s relative-entropy/modular-Hamiltonian work): the first law $\delta S = \delta \langle H_{\text{mod}} \rangle$ yields (linearized, then nonlinear) Einstein equations. The **new** step is to show the modular Hamiltonian’s flow *is* the **Godbillon–Vey stiffness** of the exotic foliation (Connes’ flow of weights = GV), so that “minimizing geometric tension” = extremizing GV = the equilibrium condition. (*Advance in hand: the flow is geometrically canonical, the favorable setting. Caveat: Faulkner’s version is holographic/emergent — in tension with Plank I; lean on the intrinsic Jacobson/Casini versions.*)

2. **Predictivity of the exotic channel.** Fix one exotic (GV) structure and derive $\{\Lambda, \text{inflation}, m_H, m_\nu\}$ from it with no further freedom; eliminate the shared modulus pairwise and test the **parameter-free relation** $\Phi(\Lambda, m_H) = 0$ against data. One confirmed relation makes the channel predictive.
3. **Complete the constructive correspondence (§5)** beyond the protected sector / beyond Brieskorn (the moduli / higher- b_1 case, e.g. $\Sigma_g \times S^1$); and constructive Osterwalder–Schrader for the interacting theory.
4. **Scholarship gate (prerequisite to any novelty claim).** This is a **synthesis of established results** organized around new framings. **What is *not* novel here (and must be credited as others’):** that entanglement equilibrium / the first law *derives* gravitational dynamics (Jacobson; Faulkner–Guica–Hartman–Myers–Van Raamsdonk; Casini); that the universal entanglement term is the sector category (Casini–Huerta–Magán–Pontello); the DR/Tannaka, special-geometry, Taubes, and Connes–GV facts. **The genuinely candidate-novel content is narrow:** (a) identifying the protecting modular structure with the **Godbillon–Vey stiffness** of the exotic foliation (and “rigidity = protection = clock”); (b) the **monopole-Floer** $\leftrightarrow$ **DHR** join. Both are unverified by machine search and **folklore-adjacent** until confirmed. Run MathSciNet/zbMATH and consult experts in the overlap — **the entanglement-equilibrium/modular community** (Faulkner, Casini, Jacobson; and Rovelli on thermal time), the **Rome AQFT school** (Vasselli, Ruzzi), **Asselmeyer-Maluga–Król**, and **SW-Floer-as-TQFT** (Manolescu). *Recommended order:* scholarship $\rightarrow$ predictivity test (2) $\rightarrow$ mechanism (1) and the constructive gaps (3).

7. Summary

Geometry-first, pursued constructively, builds a coherent picture on the **protected stratum**: geometry fixes the gauge group, the charges, the sectors, the universal entanglement, and — through special geometry — the low-energy dynamics; exotic smoothness is physical; and the protecting structure that makes all of this possible is, on the central conjecture, the **stiffness of spacetime** (Godbillon–Vey folding rigidity), which doubles as the origin of time and of the 3+1 split. The four planks are one object — the stiffness/spectral-curve of the 4-geometry — seen from different sides. The forward program is the four items of §6.

8. Glossary — routes reviewed